Mikołaj Kurek

Applied Computer Science Engineer | Edge AI, Computer Vision & Robotics | Med-tech R&D

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PROFESSIONAL PROFILE

Applied Computer Science engineer (BEng) and MSc student specialising in real-time computer vision on edge hardware, robotics and 3D-printed medical devices. As co-founder and CTO of a med-tech startup working with the National Institute of Oncology, I turn patient CT scans into 3D-printed brachytherapy applicators and automate their design with custom Python tools. I build complete systems end to end, from dataset and model training through electronics and mechanics to deployment. Currently on an Erasmus+ semester in robotics and mechatronics in France.

Research interests: efficient vision models for embedded and robotic systems, robot perception, and medical image-to-device pipelines.

TECHNICAL SKILLS

AI & Computer Vision	YOLO (training, INT8 quantisation, edge deployment), TensorFlow/Keras (CNNs), OpenCV, MediaPipe, Scikit-learn, Pandas, LLaVA (Ollama), Label Studio, Albumentations
Robotics & Embedded	Raspberry Pi 5 + AI Camera (Sony IMX500), Arduino, Siemens S7 PLCs, sensors, MOSFET drivers, I2C, KiCad (schematic & PCB), soldering
Software & Web	Python, C++ (Arduino), JavaScript/TypeScript, React, React Native (Expo), FastAPI, Flask, PostgreSQL, Supabase
Mechanical & Simulation	Fusion 360, SolidWorks, Creo, Ansys Fluent (CFD), Ansys Mechanical (FEM), GD&T, ISO fits, machine-element calculations
Fabrication & Med-tech	FDM 3D printing (2+ years, Bambu Lab X1C), slicer tuning, medical-grade polymers, DICOM-to-STL, Blender Python API
Data & DevOps	Power BI (DAX, Power Query), jamovi (ANOVA), Docker, Docker Compose, Zabbix, Nginx, Git, GitHub Actions, SQL
Languages	Polish (native), English (C1, certified)

PROFESSIONAL EXPERIENCE

Co-founder & CTO, Med-Tech Startup 01/2025 - Present

In collaboration with the National Institute of Oncology, Kraków, Poland

- **Problem:** Skin brachytherapy usually relies on hand-shaped thermoplastic masks, where catheter guides shift, air gaps remain against the skin and every case takes repeated clinician time.
- **Medical imaging pipeline:** Built the end-to-end workflow from patient CT (DICOM, Hounsfield-scale segmentation) to STL, applicator model, treatment-planning simulation and 3D print.
- **R&D & automation:** Designed and programmed a Blender Python API add-on that automatically generates and inspects catheter channels for radiation sources at a fixed distance from the skin.
- **Manufacturing engineering:** Lead 3D printing in biocompatible Spectrum Medical ABS (USP Class VI, ISO 10993-1, autoclavable); developed print parameters and shrinkage compensation for dimensional stability.
- **Clinical collaboration:** Work directly with oncologists, medical physicists and technicians; applicators delivered for the finger, both ears and the scalp, with 3 to 5 catheter channels and anatomical reference points for repeatable placement.
- **Impact:** CAD preparation time cut by over 50% and under one week from CT scan to finished applicator, within strict clinical tolerances.

IT & Industrial Systems Programmer, Dan Cake 07/2022 - 09/2022

Large industrial bakery, Chrzanów, Poland

- **Logistics optimisation:** Deployed an application that reads industrial scanner data on production lines and flags pallets that need reinforcement, eliminating losses from products crushed under load.
- **Industrial visualisation:** Built real-time monitoring of raw-material silos, tanks and line throughput, reading directly from Siemens S7 PLCs.
- **Production tools:** Line-traffic manager reconciling PLC scanner data with PC records, and a CRUD database application for a warehouse.

KEY ENGINEERING PROJECTS

AI Vision Conveyor Sorter (BEng Thesis) 2025 - 2026

YOLOv8n, INT8, Sony IMX500, Raspberry Pi 5, Python, OpenCV, KiCad, Fusion 360, Figma

- **Problem:** Quality control on small production lines needs expensive, hard-to-reconfigure PLCs and industrial cameras. Could one person build a low-cost sorter that recognises products by appearance and learns new ones in software?
- **Key constraint:** Real-time detection on a moving belt without overloading the Raspberry Pi. I ran the network on the camera sensor itself, which meant fitting the whole model into 8 MB of memory.
- **Method:** Built my own 7-class dataset (186 photos, 12x augmentation), trained YOLOv8n with transfer learning and quantised it from FP32 to INT8 (7.06 of 8 MB), so the Pi only receives finished detections.
- **Hardware:** Designed and built the T-slot conveyor, 3D-printed every custom part (Fusion 360), designed and soldered the PCB (KiCad), and wrote multithreaded Python control with a Figma operator dashboard.
- **Result:** 96% sorting accuracy on moving objects (82/85 trials), mAP@0.5 of 0.995, 46 ms inference at 30 fps, and 60 minutes of continuous loaded running with no lost steps.

ECG Anomaly Detection (Data Science) 2024

Python, TensorFlow/Keras, scikit-learn, Pandas, Flask

- **Problem:** Class imbalance. Arrhythmias are far rarer than normal heartbeats (about 83% of the MIT-BIH data is normal), so a model that always answers "normal" scores high accuracy while missing every dangerous case. New samples could not be added to balance the data.
- **Method:** 1D CNN trained with class weights, so a missed arrhythmia costs about 14 times more than a false alarm. Recall was the main metric instead of accuracy, and the decision threshold was lowered to 0.3.
- **Result:** 97.6% of arrhythmias detected (70 of 2,934 missed) at 91% accuracy. The trade-off is more false alarms, which a doctor can rule out, instead of missed diagnoses.

FORMA: AI Fitness Coach App (team of 2) 2026

React Native, Expo, TypeScript, Supabase, PostgreSQL, Google Gemini, Jest, GitHub Actions

- **Problem:** Fitness apps are either plain logbooks with no analysis or complex paid platforms. The goal was fast set-by-set logging with real analysis of the user's own training history.
- **Method:** An LLM is only useful if it judges real data, not guesses. Gemini receives a structured prompt with the user's profile and four weeks of sets, needs at least three sessions before it judges, and returns a weekly assessment, next week's plan and overtraining warnings.
- **Engineering:** One iOS/Android codebase, Supabase with row-level security, Jest tests (over 60% coverage) and CI.

NutriVision AI: Visual Meal Analysis 2025

React, FastAPI, PostgreSQL, Docker Compose, Ollama, LLaVA, Zabbix

- **Problem:** Food photos are personal data, yet most calorie apps send them to third-party cloud models.
- **Method:** Ran a vision LLM (LLaVA) locally, so images never leave the server. To keep it stable on limited RAM, images are downscaled and optimised with Pillow before inference, and Zabbix tracks container load and model response time.

EV Fleet Efficiency Analysis (Statistics) 2026

Power BI, DAX, Power Query, jamovi

- **Problem:** Fleet budgets assume the same energy use every month. Does weather change efficiency, and is it the cold or the snow?
- **Method:** Star-schema model of three years of Chicago fleet and weather data. The data broke classic ANOVA assumptions (Levene, Shapiro-Wilk), so I used Welch ANOVA with Tukey post-hoc.
- **Result:** Winter costs 53% more energy per mile ($p < 0.001$), driven by temperature rather than snow; recommended a 30 to 50% winter energy budget.

Engineering Simulation and Machine Design (Coursework) 2024 - 2025

Ansys Fluent, Ansys Mechanical, CAD, GD&T

- **Method:** A result is only as good as its mesh and checks, so I used symmetry to reduce models, mesh-refinement studies (error cut from 0.31% to 0.03%) and energy and mass balance checks (below 0.002% error); a composite plate loaded to failure at about 107 kN.
- **Machine design:** Load calculations and workshop drawing for a 12 kW gearbox shaft (stepped 35 to 45 mm, ISO fits), plus a hand wheel and a 250-bar hydraulic cylinder.

Self-Watering Plant Pot (Mechatronics, team of 2) 2025

Arduino Uno R4, C++, MOSFET, I2C LCD, SolidWorks, FDM

- **Problem and method:** Water soaks in slowly, so watering until the sensor reads "wet" overwaters. I used closed-loop pulses instead (water 5 s, wait 2 min, re-measure) with a MOSFET-switched pump, in a 3D-printed enclosure.

AI Trainer: Human Pose Estimation 2024

Python, MediaPipe, OpenCV

- **Method:** Treated motion as geometry: joint angles from 33 tracked landmarks split each exercise into phases, so a full cycle counts one repetition and out-of-range angles flag form errors.

EDUCATION

Erasmus+ Exchange: Advanced Robotics, Mechatronics and Automatic Control (ARMAC) 08/2026 - 01/2027

Université Marie et Louis Pasteur (UMLP), Besançon, France

MSc, Applied Computer Science 02/2026 - Present

Cracow University of Technology. Focus: computer vision, efficient models for edge hardware, engineering automation

BEng, Applied Computer Science 2022 - 2026

Cracow University of Technology. Thesis: AI vision conveyor sorter, defended 02/2026

Open Photography Studies 03/2024 - 06/2024

Faculty of Architecture, Cracow University of Technology. Composition, lighting and image analysis

CERTIFICATIONS

English C1	Language Centre, Cracow University of Technology (SJO PK), 02/2024
EE.09	Polish national IT qualification: programming, websites and databases, 03/2022
EE.08	Polish national IT qualification: computer systems and networks, 09/2021

ACTIVITIES AND EVENTS

NASA Space Apps 2026	Co-lead of a local event (~50 participants) of NASA's international hackathon
ASTRA FC	Member of the UMLP student club for robotics and aerospace projects (since 2026)
Universal Robots	Collaborate: international cobot event, Tauron Arena Kraków (05/2025)
HackYeah 2023	Europe's largest hackathon: built game mechanics and logic in the 24h Game Jam